OPIM639 · Innovations & Operations of Intelligent Online Marketplaces

Chagee: opening overseas stores right

One new store: forecast demand, plan the inventory, choose the counter, review

Group 3

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Six chapters: one new store, four gates from forecast to review

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About 6:55 on stage, then 3:00 for Q&A. Click a chapter to jump to it. Appendix slides are in the handout only.

One new store, no local sales history: estimate demand, plan the inventory, choose a counter, then update

We are opening a store with no local sales history. We need to estimate demand, translate that demand into an inventory plan, choose a counter configuration, and update those decisions after opening.

Why it matters for Chagee

345 overseas stores in 7 markets and about 200 openings planned a year; each commits stock and a counter before local sales exist.

Proposed AI (not built here)

Forecast the whole store's demand from comparable stores, cross-checked by pre-launch app sign-ups, and validate it against actual sales.

What this prototype calculates

Synthetic inputs: a demand range, one inventory worked example (fresh milk) and a counter simulation. Tea, other ingredients and packaging are planned qualitatively. No trained model.

Who decides

People: teams set the inventory plan and choose the counter from these outputs and local constraints.

INTERACTIVE EXPLAINER WHERE AI ENTERS A NEW STORE'S OPENING

Click a step. Today's practice is our reading of public sources and one industry conversation, not a Chagee statement.

1 Members → 2 Forecast → 3 Opening inventory → 4 Counter → 5 After opening

STEP	TODAY (OUR READING)	WHAT AI ADDS	WHO DECIDES · WHERE
1 · Members	Members order through the app (our understanding).	**Read sign-ups and pre-orders as early demand evidence.**	Membership / CRM *slide 3*
2 · Forecast	Judgement-led by a local General Manager (our reading).	**A proposed demand range from member signals, with hype and habit separated by existing-market proxies.**	Market-entry leads and GMs *slide 3*
3 · Opening inventory	Direct, standardised sourcing of tea and milk for domestic stores (company release).	**Turn the forecast into an item-by-item opening list; size fresh milk with newsvendor logic as the worked example.**	Supply chain / planning *slides 4 and 5*
4 · Counter	Moving toward semi-automation: QR-scan dispensing, Chagee Express.	**Simulate the counter before the store exists, and flag which inputs would change the choice.**	Store format / equipment; franchisees *slides 6 and 7*
5 · After opening	Not covered by public sources.	**Compare actual demand with the range to narrow the next market's forecast.**	Market-entry leads *slide 8*

Printed with all five steps shown. Live: click a step to see one at a time.

FOUR GATES

Predict → Provision → Counter → Review.

Scope: the whole store. Each gate states the decision, its evidence, a suggested action, the trade-off and what the model cannot establish. Figures are illustrative assumptions.

A closed app-to-store channel puts our problem in Topics 3 to 5, not the marketplace topics

INTERACTIVE EXPLAINER EIGHT COURSE TOPICS: CLICK ONE TO SEE WHY IT IS IN OR OUT

TOPIC 1
~~Introduction to Online Retailing~~
OUT
One closed channel, app to store: no online-versus-store trade-off to weigh (our reading).

TOPIC 2
~~Online Marketplaces~~
OUT
No third-party sellers, so no sell-versus-fulfilment-service choice (our reading).

TOPIC 3
Analytics and AI for Online Retailing
IN
→ Predict · slide 3
Predicting a new market's demand from app-member signals.

TOPIC 4
Supply Chain Management using AI and Robotics
IN
→ Provision · Counter
Supply-chain planning turns the forecast into stock; equipment selection sizes the counter.

TOPIC 5
Order Fulfillment for Online Retailing
IN
→ Provision · Counter
Preparing the opening inventory, and work shared by people and robots (the bucket brigade).

TOPIC 6
Last-mile Delivery for Online Retailing
NOT USED
Outside our question, which stops at the counter.

TOPIC 7
Service-oriented platforms
NOT USED
Outside our question.

TOPIC 8
~~Omni-channel Retailing~~
OUT
One channel, so no channel conflict; the franchise revenue share is out of scope.

INTERACTIVE EXPLAINER WHERE DOES UNCERTAINTY LAND?

Why this slice: with sourcing fixed, a demand miss lands in the store, on its opening stock or its counter speed (our reading, not a Chagee statement).

Rigid sourcing | Localised sourcing

Rigid sourcing — Chagee today, our reading		Localised sourcing — For contrast only	
Sourcing ○○○	Fixed: single origin	**Sourcing** ●●○	Flexes with local suppliers
Distribution ○○○	Fixed: run from HQ	**Distribution** ●○○	Shorter, local
Opening stock ●●●	Absorbs the miss	**Opening stock** ●○○	Less exposed
Counter speed ●●●	Absorbs the miss	**Counter speed** ●●●	Still exposed

●○○ low ●●● high exposure to a demand miss

Printed: every tile shows its status and reason; both sourcing columns are shown.

OUR SCOPE

Three topics, three decisions.

Topic 3 feeds Predict; Topics 4 and 5 feed Provision and Counter. Sourcing is fixed by design, so we take it as given, and the franchise revenue share stays out of scope.

Predict: plan on a demand range, not one number, and treat the range as an assumption to test

GATE 1 · PREDICT

DECISION

What demand should the whole store's inventory and counter planning assume?

SUGGESTED ACTION

Plan on about 3,889 cups a week, within an illustrative 80% range of 2,625 to 5,153; treat it as an assumption to test.

EVIDENCE

Base **3,889 cups/week** 80% range **2,625–5,153**

Half-width **±32.5%**

TRADE-OFF

Plan too low: stockouts, queues and lost sales. Plan too high: stock that expires or ties up cash, and machines or staff that sit idle.

LIMITATION

The range is assumed, not calibrated, and demand can fall outside it. It cannot approve a site or show that a store is viable.

Demand prediction `COURSE CONCEPT · TOPIC 3`

`PROTOTYPE: AN ASSUMED RANGE, NO TRAINED FORECAST` Estimate how much customers will buy, as a range rather than one number. *Informs: the whole store's inventory and counter planning*

Proposed AI: comparable stores first, sign-ups as a cross-check

A model trained on Chagee's existing stores would set the base; app sign-ups would cross-check it. Not built here: our base is assumed.

A narrower range is an assumption to test

The slider narrows it from ±50% to ±15% by assumption; no evidence yet that member data does this.

Hype versus habit needs a proxy

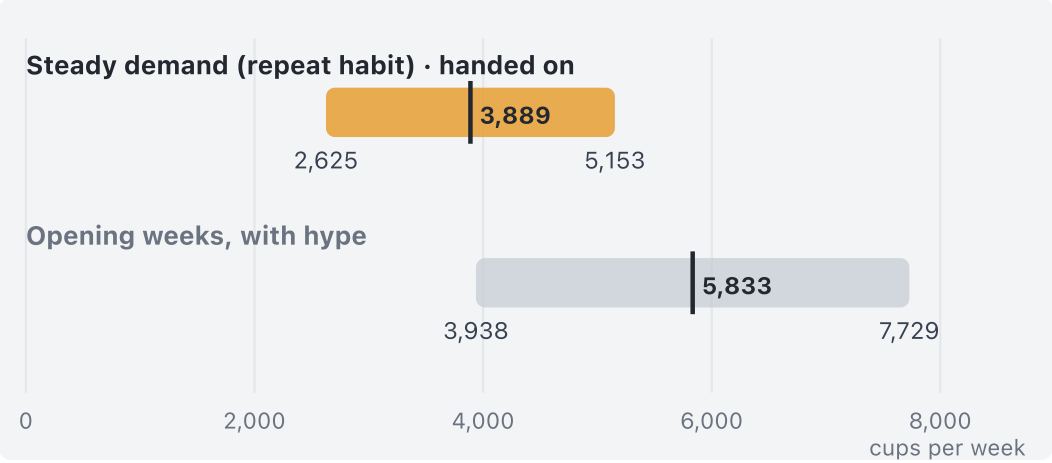
No repeat history yet: use repeat rates in existing markets or nearby stores. Kuala Lumpur reportedly sold 1,000+ cups on day one (secondary).

Blind spot: tourists and non-app orders

They sit outside the member signal, so the range runs low where tourists matter.

INTERACTIVE EXPLAINER DEMAND RANGE BEFORE OPENING: A SENSITIVITY

Weekly cups at one new store. The bar is an illustrative 80% demand range: we assume about that share of outcomes falls inside it.



Assumed signal strength — 50% → ±32.5%

Opening hype removed | Hype included

Plan on about 3,889 cups a week of steady demand (opening hype removed). Illustrative 80% demand range: 2,625 to 5,153. The range is assumed, not calibrated; demand can fall outside it.

Printed at the default assumed signal strength (50%) with opening hype removed.

WHAT THE RANGE IS FOR

Low end (2,625)

Risk of excess stock (perishables expire) and unused capacity.

High end (5,153)

Risk of shortages, queues and lost sales.

Provision turns the range into the opening inventory list; fresh milk adds shortage and waste costs.

Counter uses demand, uncertainty and local wages and machine costs together.

Not in the signal: tourists and non-app orders.

Provision: translate the sales forecast into an opening inventory plan

GATE 2 · PROVISION

DECISION

What must be on hand on opening day, what should the first delivery bring, and what is replenished later?

SUGGESTED ACTION

Build the opening list item by item from the demand range: tea, milk and alternatives, other ingredients, cups, lids and packaging. Size fresh milk with the worked example.

EVIDENCE

Quantified here **`fresh milk only (slide 5)`**

Needs Chagee data **`menu mix · recipes · lead times · minimum orders`**

TRADE-OFF

Too little of any one item stops sales. Too much ties up cash and storage, and perishables expire or lose quality.

LIMITATION

No quantities for tea, other ingredients or packaging: their inputs are not in our sources. Inventory and counter are not optimised together.

FROM FORECAST TO PURCHASE ORDER

1. **Forecast cups**
   The whole store's weekly range (slide 3). Cups alone are not enough: drinks use different ingredients.
2. **Product mix**
   Share of each drink and format (size, hot or cold). Requires menu-mix data.
3. **Recipes and materials**
   Ingredient requirement = forecast drink quantities × ingredient amount per drink. Packaging follows orders and formats.
4. **Deliveries and stock**
   The order nets off usable stock and deliveries due, adds expected losses, and fits lead times, shelf life, storage and supplier minimums.

Preparing inventory for fulfilment

`COURSE CONCEPT · TOPIC 5` `PROTOTYPE: FRESH MILK ONLY`

Have every material on hand before the orders arrive, item by item. *Informs: what must be on hand on opening day*

Lead time `EXPLAINED, NOT MODELLED` The time from placing an order to having it usable in the store. *Informs: how early each item is ordered*

Reorder point and safety stock `EXPLAINED, NOT MODELLED` Reorder when stock falls to the usage expected during the lead time, plus a buffer for demand above it. *Informs: when each item is replenished*

OPENING PLAN BY CATEGORY: QUALITATIVE, NO QUANTITIES

ITEM	WHAT DRIVES ITS DEMAND	MAIN CONSTRAINT	OPENING DECISION	REPLENISHMENT	NEEDED FOR A NUMBER
Tea leaves	Tea drinks in the mix × tea per drink	Sourcing lead time, minimum orders, storage; quality fades with age and handling, so tea also has waste and quality risk	How much to hold before opening: the opening weeks plus the time a reorder takes to arrive	Reorder when stock reaches usage over the lead time plus a buffer	Recipes · supplier lead time · minimum order · usable life
Fresh milk and alternatives	Milk drinks × milk per drink; alternatives by customer choice	Short shelf life, chilled space, delivery frequency	First fresh-milk delivery: the worked example (slide 5)	Each delivery sized on actual sales	Recipes · share of alternative milks · delivery schedule
Other ingredients	Each recipe's amounts	Each has its own shelf life and supplier terms	Plan per recipe and shelf life: perishables like milk, longer-life items like tea	Per item's usage and lead time	Menu recipes · shelf lives · supplier terms
Cups, lids, seals, packaging	Orders by format	Lead time, minimum orders, bulky storage	Cover opening demand plus the reorder lead time: no cup or seal, no sale, even with ingredients ready	Reorder on usage and lead time	Format mix · supplier lead time · storage space

Origin (imported or local) is not public for overseas stores: Chagee's direct sourcing of tea and milk is documented for its domestic operations (L-prn). Quantities, lead times, shelf lives and service levels are not in our sources, so none are shown.

MATERIALS AND CAPACITY

A forecast becomes sales only if the store has both materials and service capacity.

Enough milk is not enough if tea or cups run out. Enough ingredients are not enough if the counter cannot serve customers (slides 6 and 7). Both use the Predict range; the prototype does not optimise them together.

Worked example: choosing the first fresh-milk delivery

The whole-store plan includes many materials. We use fresh milk to demonstrate the ordering trade-off because unused stock can expire quickly.

GATE 2 · PROVISION · FRESH MILK

DECISION

How much fresh milk should the first delivery bring?

SUGGESTED ACTION

Steady demand: about 788 L for the first 7 days. If the expected opening surge is real, about 1,182 L: this delivery falls inside it.

EVIDENCE

Short vs waste, per L **S$25.47 vs S$2.55**

Order covers **91% of outcomes** Expected waste **209 L**

Expected shortage **42 cups**

TRADE-OFF

Order less: more missed cups at S$3.82 each. Order more: more milk thrown away at S$2.55 a litre. A shortage costs more, so the order sits above base demand.

LIMITATION

Sizes one delivery of one item; it does not optimise the whole inventory. Still runs short in about 9% of outcomes. Generic milk costs, one amount per cup.

Newsvendor logic `IMPLEMENTED FOR FRESH MILK` Choose the order quantity by balancing the cost of unused stock against the cost of running short. *Informs: the size of the first fresh-milk delivery*

Decision and method

How much milk should the first delivery bring? Newsvendor logic balances the cost of leftover stock against the cost of running short.

Inputs

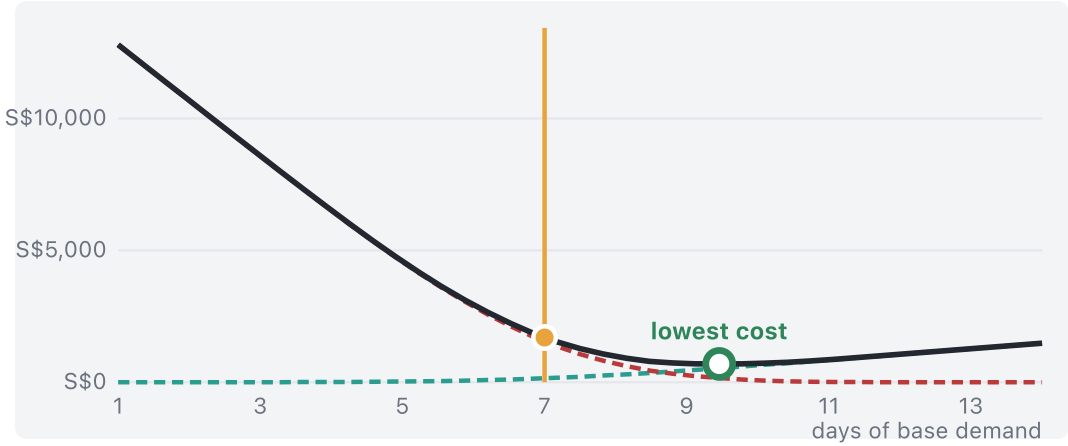
The demand range from slide 3; a missed cup costs S$3.82, a wasted litre S$2.55; shelf life about 7 days. Simplification: every cup uses 0.15 L; real recipes differ.

Opening surge or steady demand

The surge is assumed to last 2 weeks, so this delivery falls inside it: about 1,182 L if the surge is real, 788 L on steady demand.

INTERACTIVE EXPLAINER **FIRST FRESH-MILK DELIVERY CALCULATOR**

Expected cost of the first delivery at one store, using the range from slide 3. Milk costs are generic: overseas sourcing is not public.



Total expected cost · Milk wasted · Cups missed

Days of opening stock 7 days

Order about 788 L for the first 7 days, using steady demand (opening hype removed). That is 9.5 days of base demand, just above the top of the range (top of range 9.3 days).

This delivery falls inside the assumed 2-week opening surge (×1.5): if the surge is real, order about 1,182 L, or include it on slide 3.

AT 7 DAYS OF STOCK

S$1,695 expected

Milk wasted **S$149**

Cups missed **S$1,546**

BEST ORDER AT THIS BAND (±32.5%)

9.5 days **S$694**

Same order, strong signal **S$320**

Hand-off: fresh milk is one line of the opening list (slide 4). The counter is decided by volume and input costs, not this band (slide 7).

Printed at the band from slide 3 (opening hype removed) and the default of 7 days of stock.

Counter flow: people pass work along like a bucket brigade; machines are fixed, so they queue at the bottleneck or sit idle

Bucket brigade `COURSE CONCEPT · TOPIC 5` `ILLUSTRATION WITH THE WEEK 5 RULES` Workers pass each order along a line and rebalance themselves, slowest first. *Informs: how much a crew absorbs in a surge*

Bottleneck `COURSE CONCEPT · TOPIC 5` `ILLUSTRATION AND COST MODEL` The slowest step sets the pace of the whole line. *Informs: where a fixed machine line queues*

People pass the work along

Each worker carries a drink downstream and hands it over. In a surge the handover point moves, so the load evens out: one way a crew shares work.

Machines stay at their station

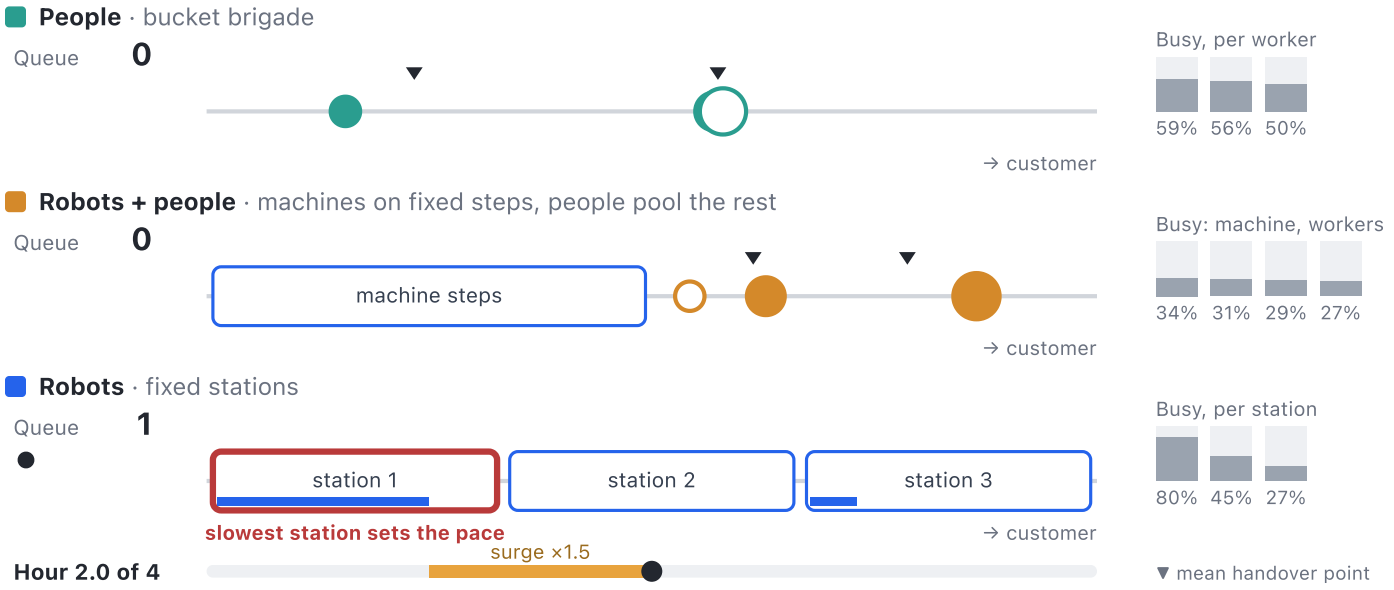
A fixed station cannot hand work back or take on more. The queue builds at the slowest one, wherever it sits on the line.

Faster in the peak, idle at low volume

Machines can cut the wait in a surge, but sized for the peak they are busy only part of the day, and idle capacity is what a store pays for.

INTERACTIVE EXPLAINER COUNTER FLOW: PEOPLE AND MACHINES IN A SURGE

Movement illustration in Week 5 units: not a cost model, not measured times. More detail: appendix A5.



Moderate ×1.5 surge, 4-hour run	Machines matched to crew		As sized in cost model	
	Peak queue	Busy (range)	Peak queue	Busy (range)
People	0	57%–66%	0	57%–66%
Robots + people	0	28%–32%	0	10%–33%
Robots	2	38%–90%	1	19%–40%

Printed at the end of the surge window; the table compares both sizing settings over the full run.

OUR VIEW

Materials are half of opening; service capacity is the other half.

People flex; machines come in whole units. So the counter choice turns on volume and local costs: next slide.

Counter: at our inputs all-human and mixed are S$0.02 a cup apart, so choose with all inputs together

GATE 3 · COUNTER

DECISION

Which counter should this store install: all-human, mixed, or full automation?

SUGGESTED ACTION

At 556 cups a day all-human is cheaper by S$0.02 a cup: close, but above the S$0.01 tie line. Choose using all current inputs together.

EVIDENCE

Cost per cup **all-human S$0.65 · mixed S$0.66 · all-robot S$0.91**

All-robot machines busy **21% of open hours**

TRADE-OFF

All-human: cheapest here, but depends on trained staff. Machines: faster in the peak, but idle and costly when volume is low.

LIMITATION

Small gap on a simplified staffing model that misses its Week 5 check (4.7% and 6.8% against ±3%). No measured service times: a candidate to test in a pilot, not an instruction to install.

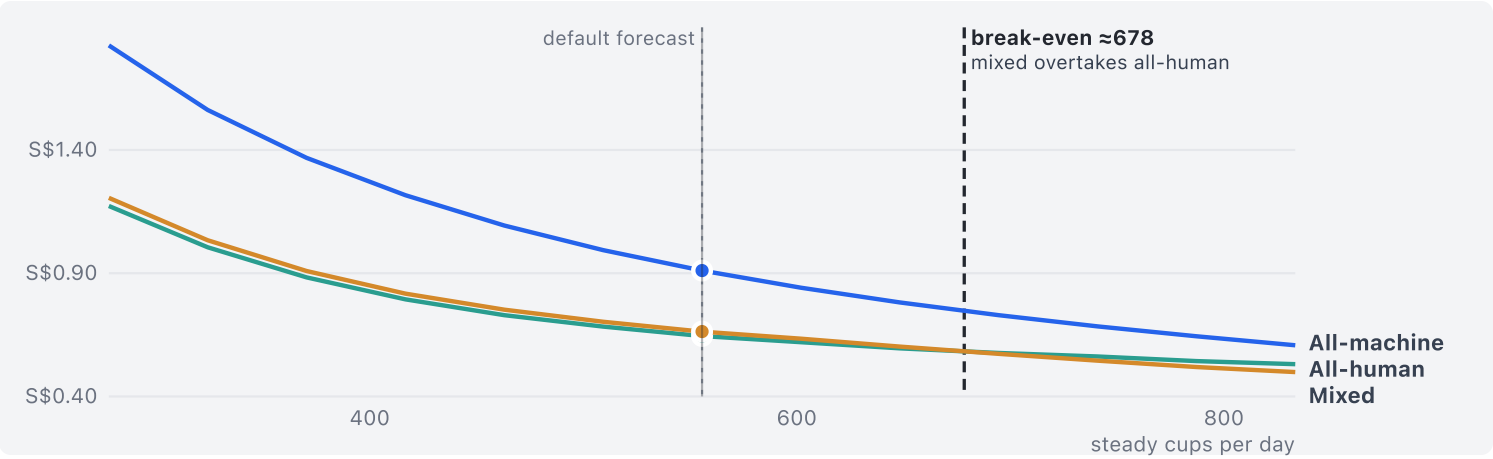
INTERACTIVE EXPLAINER COST PER CUP, BY COUNTER

Faster service does not mean a lower cost per cup: at low volume, machine capacity sits unused. Labour, machines and lost sales per cup served, by steady volume.

Capacity utilisation `IN THE COST MODEL` The share of available capacity that is actually used. *Informs: why idle machines raise the cost per cup*

Selecting robotics equipment `COURSE CONCEPT · TOPIC 4` `COMPARED IN THE COST MODEL` Choose automation that fits the volume and the work, not the most automation. *Informs: which counter to install*

All-human Mixed All-machine



All-human	Mixed	All-machine
S$0.65	**S$0.66**	**S$0.91**
cheapest	+S$0.02	+S$0.27

At 556 cups/day. All-robot machines are busy 21% of open hours.

Printed at the default forecast.

ONE-VARIABLE THRESHOLDS: WHICH ASSUMPTIONS DESERVE A CLOSER LOOK

MIXED WINS WHEN…	IS	OURS	GAP
Machine cost	below **S$1,800/month**	S$1,909/month	-6%
All-human coordination	below **0.85**	0.90	-6%
Wage per hour	above **S$15.97/h**	S$15.06/h	+6%
Mixed coordination	above **0.98**	0.90	+9%
Steady volume	above **678 cups/day**	556 cups/day	+22%
Forecast band	above **±57%**	±32.5%	+75%
New-hire share	none in 0% new staff to 50% new staff: all-human cheaper throughout		
Machine time	none in 0.70× to 2.00×: all-human cheaper throughout		

Each row moves one input with the rest at our defaults; nearest first. A crossed row is a reason to recheck, not a rule to switch.

How sure are we? A close call. The gap is S$0.02 a cup. Our staffing shortcut misses its Week 5 check (4.7% and 6.8% against ±3%), and with walking modelled people reach only 90% of the assumed capacity (70% at today's counter). Test the ranking with real staffing data before relying on it.

Review: rerun the comparison with observed inputs and compare it with the original plan

GATE 4 · REVIEW

DECISION
After real sales arrive, does the plan still hold?

SUGGESTED ACTION
Create an updated scenario using observed inputs and compare it with the original plan (All-human by S$0.02 a cup).

EVIDENCE
Feeds the model **steady sales · wage · machine quote · milk costs**
Check by hand **service times · lost orders · waste and stockouts of every item vs plan**

TRADE-OFF
Change too early and you chase the opening hype; change too late and the store keeps the wrong stock or counter.

LIMITATION
Rerunning the model does not validate it, and a new result does not by itself prove that machines should be added.

TRIGGER	FLIPS WHEN	OURS	GAP
Counter: when mixed beats all-human · Store format / equipment, with franchisees			
Machine cost	below **S$1,800/month**	S$1,909/month	-6%
All-human coordination	below **0.85**	0.90	-6%
Wage per hour	above **S$15.97/h**	S$15.06/h	+6%
Mixed coordination	above **0.98**	0.90	+9%
Steady volume	above **678 cups/day**	556 cups/day	+22%
Forecast band	above **±57%**	±32.5%	+75%
New-hire share, Machine time	no threshold in the tested range: all-human stays cheaper throughout		
Counter: when full automation beats mixed · Store format / equipment, with franchisees			
Steady volume	above **1,820 cups/day** · step	556 cups/day	+228%
Wage per hour, Machine cost, Forecast band, Machine time	no threshold in the tested range: mixed stays cheaper throughout		
Fresh milk: when to order at or below the base forecast · Supply chain / planning			
Milk waste per litre	none in S$1.27 to S$5.09/L: order above base throughout (flips at S$25.47/L = S$3.82 per lost cup ÷ 0.15 L)		

One input at a time, the rest at our defaults; nearest first. Thresholds flag what to recheck, not when to switch (slide 7).

INTERACTIVE EXPLAINER — **UPDATED SCENARIO VERSUS THE ORIGINAL PLAN**

Enter observed inputs. Verdict: the full cost model; under S$0.01 a cup is a tie. Nearest threshold: a live estimate.

Wage per hour		S$15.06
Machine cost, per unit per month		S$1,909
Expected steady cups per day		556
Forecast band (±)		±32.5%
Milk waste cost per litre		S$2.55

Original plan: All-human by S$0.02 a cup. **Updated scenario:** same inputs as the plan.

COUNTER **All-human** cheaper than Mixed by S$0.02 a cup

All-human **S$0.65** Mixed **S$0.66** All-machine **S$0.91**

Nearest threshold (live estimate): Mixed would be cheaper if machine cost, per unit per month falls to about **S$1,783** (7% away).

Fresh milk: order above the base forecast (best order at the 91% point of demand); flips only if waste costs over S$25.47 a litre.

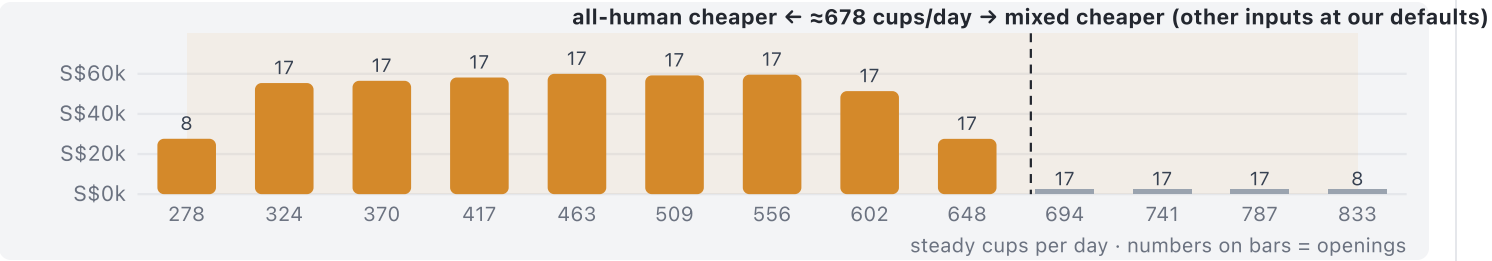
Printed at our default inputs.

Each opening gets a demand estimate, an inventory plan, a counter tested on local costs and a review; only counter and fresh-milk benefits are quantified

INTERACTIVE EXPLAINER WHAT CHOOSING PER STORE IS WORTH (QUANTIFIED PART ONLY)

Comparison: a mixed counter at every opening versus choosing each store's counter for its modelled conditions. The quantified inventory benefit currently covers the fresh-milk example only. Benefits for tea, other ingredients, and packaging have not been estimated.

Bar = openings at that volume × the standard's yearly cost above the cheapest counter (our defaults).



Overseas openings a year	200
Expected steady volume: lowest	278 cups/day
Expected steady volume: highest	833 cups/day

RECURRING · COUNTER COSTS
S$455k
a year, for one year's openings

ONE-TIME · FRESH-MILK STOCK
S$74k
once; other inventory not estimated

142 of 200 openings sit where all-human is cheaper → S$455k a year; **S$370** per opening × 200 → S$74k once

Stock term: fresh milk only, and only if the pre-launch signal works as assumed (slide 3). A fully automated standard would cost S$32,724–S$65,190 more per store a year.

Printed at our defaults: openings spread evenly over our volume range.

FOUR OUTPUTS FOR EACH OPENING

1. **Before committing the site** · Predict `slide 3`
   A demand estimate for the whole store: a range to test, back-tested on past openings.
   Market-entry leads · Membership / CRM
2. **Before the first order** · Provision `slide 4`
   An opening inventory and replenishment plan, item by item; fresh milk sized with newsvendor logic.
   Supply chain / planning
3. **Before fit-out** · Counter `slide 7`
   A counter compared on demand and local costs, all inputs together; a candidate to pilot.
   Store format & franchisee
4. **After opening** · Review `slide 8`
   Compare sales, stockouts, waste and service with the plan; rerun with observed inputs.
   Store operations · Market-entry leads

FINDING

In the tested model, uncertainty did not determine the counter choice.

Volume and operating costs mattered more; demand uncertainty still matters for inventory. This holds for the scenarios we tested, not as a general rule. Next: replace the assumptions with Chagee data and back-test on recent openings such as Korea (Q2 2026).

RISKS AND HOW WE WOULD MANAGE THEM

Staff dependence and turnover
Keep SOP training and flexible roles.

Opening crowds read as steady demand
Decide the counter on steady volume, after the hype.

Tourist and non-app demand is invisible
Widen the band in tourist areas; log walk-ups.

Automation versus the premium teahouse
Keep people at finishing and hand-over.

Every assumption, with its unit, range and source

ASSUMPTION	VALUE	UNIT	RANGE	SOURCE
Facts				
stores end 2025	7,453	stores	–	L-jiemian
overseas stores	345	stores	–	L-jiemian
overseas markets	7	markets	–	L-jiemian
gmv 2025 rmb bn	31.58	RMB bn	–	L-jiemian
net revenue 2025 rmb bn	12.91	RMB bn	–	L-jiemian
tea latte share china	0.91	share of China sales	–	L-prometheus
kl opening day cups	1,000	cups (reported, '1,000+')	–	L-gabgrowth
Predict				
signal strength default	0.5	share, 0 = no pre-launch signal, 1 = strong signal	0 to 1	assumption
Predict Toggles				
remove hype default	yes	boolean	–	assumption
Store				
open hours per day	12	hours/day	–	assumption
open days per year	360	days/year	–	assumption
cups per store per year	200,000	cups/store/year	100,000 to 300,000	assumption
hourly demand share	0.04, 0.06, 0.1, 0.11, 0.08, 0.08, 0.09, 0.1, 0.1, 0.09, 0.08, 0.07	share of daily cups per open hour (10:00 to 22:00), sums to 1	–	assumption
Demand				
uncertainty definition	half_width_pct	text	–	assumption
band coverage	0.8	probability	–	assumption
demand floor share of base	0	share of base	–	assumption
uncertainty no signal	0.5	half-width, share of base	0.35 to 0.7	assumption
uncertainty strong signal	0.15	half-width, share of base	0.05 to 0.25	assumption
opening hype multiplier	1.5	x steady-state demand	1 to 2.5	assumption
opening hype weeks	2	weeks	1 to 6	assumption
uncertainty sweep	from 0 · to 0.6 · step 0.05	half-width, share of base	–	assumption
Surge				
surge multiplier	none 1 · moderate 1.5 · large 2.5	x forecast arrival rate	–	assumption
surge hours	2	busiest forecast hours that get the surge	–	assumption
surge default	moderate	text	–	assumption

Source "assumption" = our illustrative choice; ledger IDs are listed in A3. Ranges drive the sliders and the sensitivity chart in A2.

Every assumption, with its unit, range and source (continued)

ASSUMPTION	VALUE	UNIT	RANGE	SOURCE
Stations				
list	ticket, cup_ice, tea_milk, topping_finish, shake_seal, handover	text	–	assumption
human seconds per cup	ticket 5 · cup ice 10 · tea milk 15 · topping finish 12 · shake seal 10 · handover 8	seconds of one average worker per cup	0.8 to 1.25	assumption
machine seconds per cup	ticket 2 · cup ice 8 · tea milk 10 · topping finish 15 · shake seal 8 · handover 6	seconds per cup per machine unit	0.7 to 2	assumption
mixed machine stations	ticket, tea_milk, shake_seal	text	–	assumption
machine speed variation cv	0	coefficient of variation	–	assumption
People				
human speed variation cv	0.25	coefficient of variation, minute to minute	0.1 to 0.4	assumption
worker speed spread	0.2	+/- share around average speed	0 to 0.4	assumption
coordination loss human	0.9	multiplier on pooled throughput	0.75 to 1	assumption
coordination loss mixed	0.9	multiplier on human-pool throughput in the mixed setup	0.75 to 1	assumption
new staff share	0.25	share of roster in first months	0 to 0.5	assumption
new staff relative speed	0.7	x average speed	0.5 to 0.9	assumption
min staff on shift	all human 2 · mixed 1 · all machine 0	workers per open hour	–	assumption
all machine attendant per open hour	1	workers per open hour	0.5 to 2	assumption
staffing flexibility	0.7	share of roster gap closed after demand is known	0.3 to 1	assumption
Costs				
labour cost per hour	15.06	SGD per hour worked, fully loaded (statutory costs)	14.09 to 25.45	L-mom-pwm, L-mom-ows, L-cpf, L-sdl, L-mom-leave, L-mom-ph
machine cost per unit per month	1,909	SGD per station machine unit per month	636.2 to 3,818	assumption
lost order cost	3.82	SGD per lost order	1.91 to 7.64	assumption
capacity safety margin	1.2	x forecast peak rate	1 to 1.5	assumption
Service				
target wait minutes	5	minutes	2 to 10	assumption
Provision				
milk litres per cup	0.15	litres per cup	0.1 to 0.25	assumption
milk waste cost per litre	2.55	SGD per litre thrown away	1.27 to 5.09	assumption
milk shelf life days	7	days	3 to 14	assumption
stockout cost per cup	3.82	SGD per cup not served for lack of stock	1.91 to 7.64	assumption
opening stock days	7	days of forecast base demand	1 to 14	assumption
Benefit				
overseas stores per year	200	stores/year	50 to 300	L-jiemian
saving per cup	from_simulator	SGD/cup	–	assumption
waste avoided	from_provision	SGD/store/year	–	assumption

Source "assumption" = our illustrative choice; ledger IDs are listed in A3. Ranges drive the sliders and the sensitivity chart in A2.

Every assumption, with its unit, range and source (continued)

ASSUMPTION	VALUE	UNIT	RANGE	SOURCE
Simulation				
monte carlo draws	500	demand draws per uncertainty point	–	assumption
base seed	639	integer	–	assumption
minute step	1	minutes	–	assumption
live draws	20	simulated stores per live estimate	–	assumption
Triggers				
volume trigger range cups per day	277.8, 3,000	cups per store per day	–	assumption
Illustration				
worker speeds	1, 1.5, 2	lines per hour	–	assumption
walk back speed	6	lines per hour	–	assumption
base load share	0.6	arrival rate as share of the crew's nominal capacity (sum of speeds)	–	assumption
machine sizing	matched station rates = 3 x worker speeds (each station does a third of the work; same total speed as the cr…	text	–	assumption
mixed sizing	matched machine block rate = the people part's nominal rate (crew speed / hand share), a balanced mixed line…	text	–	assumption
machine buffer capacity	2	orders between consecutive machine stations	–	assumption
run hours	4	simulated hours per run	–	assumption
surge window hours	1, 2	start and end hour of the surge within a run	–	assumption
Validation				
breakeven	tol cost sgd per cup 0.01 · tol x share of range 0.005 · grid points 13	mixed	–	assumption
week5 comparison	tolerance pct 3 · engine Brigade() read from the ENGINE string in week5/build_opim639_deck.py, not modified …	mixed	–	assumption
cost per cup aggregation	ratio_of_sums	text	–	assumption
peak utilisation definition	forecast peak-hour arrival rate x surge multiplier, divided by the setup's nominal capacity in that hour, at…	text	–	assumption
crossover definition	smallest u at which mixed cost/cup <= all-machine cost/cup and stays <= for every larger u in the sweep, lin…	text	–	assumption
crossover stability	seeds 20 · min seeds with crossover 18 · max spread pct points 10	mixed	–	assumption
sensitivity	one-at-a-time: each parameter with a non-null range set to its low then high value, all others at default; r…	text	–	assumption

Source "assumption" = our illustrative choice; ledger IDs are listed in A3. Ranges drive the sliders and the sensitivity chart in A2.

Every assumption, with its unit, range and source (continued): business case

ASSUMPTION	VALUE	UNIT	RANGE	SOURCE
Business Case				
horizon years	5	years	–	assumption
opening year share	0.5	share of a year a store trades in its opening year	–	assumption
specialist day rate	1,000	SGD per person-day	–	assumption
working days per month	21	days/month	–	assumption
build people	4	people	–	assumption
build months	6	months	–	assumption
integration one time	250,000	SGD, one time	–	assumption
maintenance people	2	full-time people, every year	–	assumption
platform per store month	50	SGD per store in operation per month	–	assumption
review days per opening	1	analyst person-days per new store	–	assumption

Source "assumption" = our illustrative choice; ledger IDs are listed in A3. Ranges drive the sliders and the sensitivity chart in A2.

The model is stable across seeds but misses its Week 5 check, and a handful of inputs decide all-human versus mixed

Week 5 check: pooled staffing formula vs the Week 5 engine

WORKER SPEEDS	ENGINE, LINES/H	POOLED	GAP
1, 1.5, 2	4.300	4.500	**4.65%**
1, 2, 3	5.620	6.000	**6.76%**

Fails the pre-registered ±3% tolerance: the pooled formula overstates a real bucket brigade. The coordination factor (0.9) absorbs part of this.

Seed stability

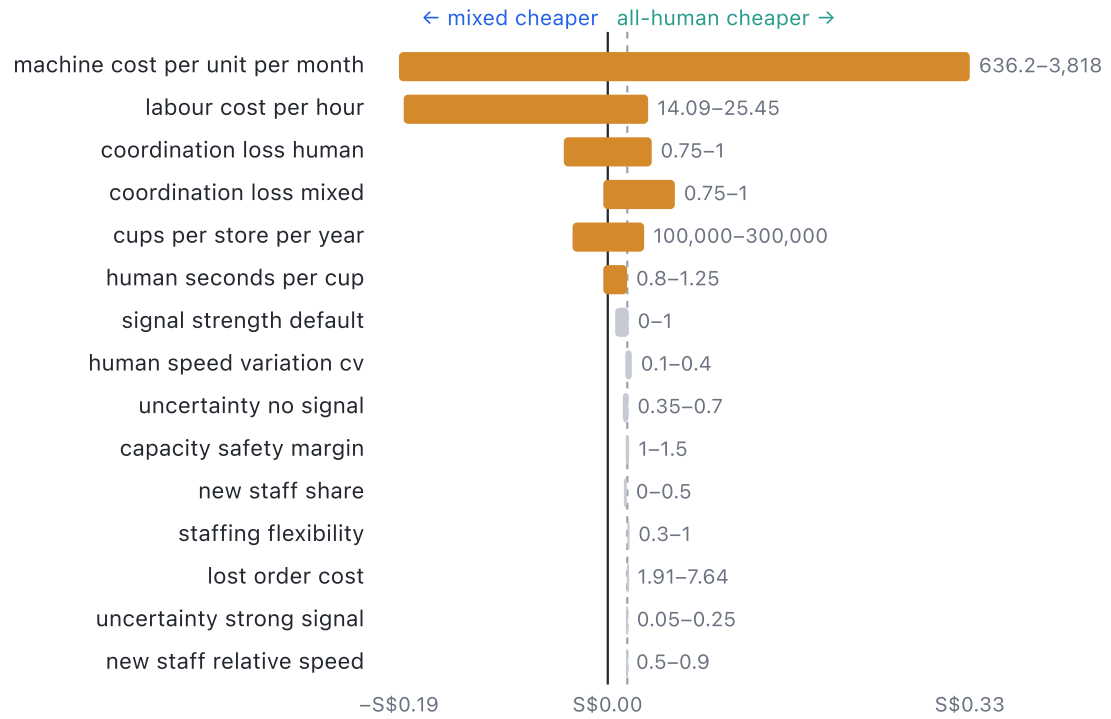
Uncertainty sweep at the moderate surge: 20 of 20 seeds: mixed cheaper than full automation at every band width. The result agrees in every seed. The pre-registered rule asks for a crossover point in most seeds; with none in any seed, the deck shows no crossover point, as the rule requires.

Store replica (walking modelled) vs cost-model capacity, cups per minute

COUNTER	REPLICA	COST MODEL	SHARE
all-human	1.50	1.67	90%
mixed	1.17	1.67	70%
all-machine	3.98	4.00	100%

The replica animates movement only; times and costs on the slides come from the cost model.

Sensitivity: mixed minus all-human cost per cup, one assumption at a time



Dashed: our default gap (S$0.018 a cup, all-human cheaper). Amber bars cross zero inside their approved range, so they can flip the choice; labels show the range tested. No visible effect: target wait minutes, worker speed spread, machine seconds per cup, opening hype multiplier, opening hype weeks, all machine attendant per open hour, milk litres per cup, milk waste cost per litre, milk shelf life days, stockout cost per cup, opening stock days, overseas stores per year.

At a new store's volume, a mixed people-and-machine counter beats full automation, because machines sit idle, not because demand is uncertain

We tested the hypothesis that mixed wins only until demand becomes predictable. Our simulator rejected the uncertainty part.

Machines come in whole units, so they sit idle

Even one machine per station serves about 240 cups an hour against a forecast peak of about 61. The fully automated line is busy only 21% to 22% of open hours.

People flex; machines cannot be returned

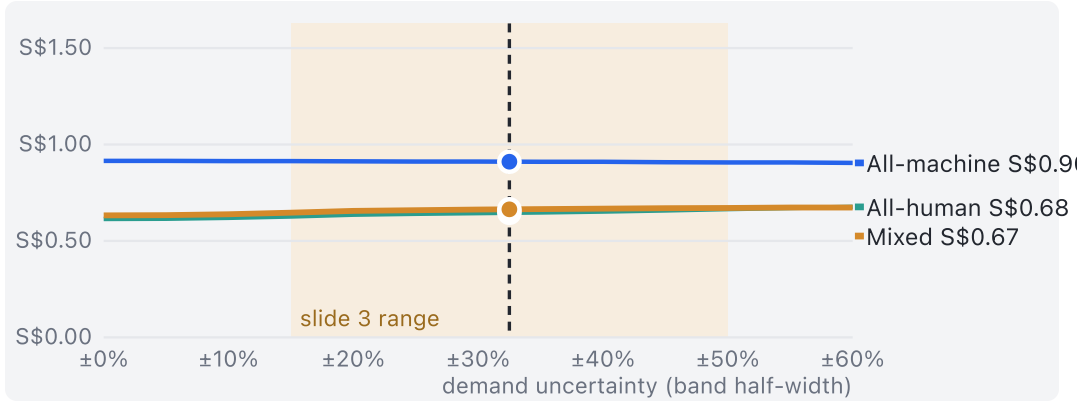
Rosters follow the hour and the real demand, and staff rebalance across stations like a bucket brigade. A mixed counter buys 3 machines instead of 6, with similar staff hours.

Where machines still win

Big surges: at a 2.5× peak, people-based counters lose about 11% of orders and full automation is cheaper. Also steady high volume, consistency and labour scarcity.

INTERACTIVE EXPLAINER COST PER CUP BY COUNTER SETUP

None ×1 surge | **Moderate ×1.5 surge** | Large ×2.5 surge



No crossover: mixed is cheaper than full automation at every uncertainty level shown.

Demand uncertainty ±32.5%

COST PER CUP AT ±32.5%

All-human	S$0.65
Mixed	S$0.66
All-machine	S$0.91
Mixed – machine	–S$0.25

WHAT THE COST IS MADE OF



Automated line busy	21%
Orders lost: mixed	1.3%

What could shift the balance

Machine cost per month	Low S$636	**Default**	High S$3,818	→ mixed cheaper everywhere
Staff turnover (share new)	Low 0%	**Default**	High 50%	→ mixed cheaper everywhere
Forecast narrows (slide 3 band)	No signal ±50%	**Slide 3**	Strong ±15%	

Printed at the default band (±32.5%) and moderate surge.

OUR VIEW

Uncertainty did not decide the counter; volume, wage and machine cost do.

All-human (S$0.65) and mixed (S$0.66 per cup at the default band) are a close call settled by the triggers on slide 7. Machine breakdowns are not modelled, which flatters automation.

The source behind every Chagee fact, and the claims we deliberately did not make

ID	SOURCE AND WHAT IT SUPPORTS	STATUS
L-jiemian	Jiemian (en.jiemian.com), earnings-call coverage: 7,453 stores end-2025; about 200 overseas stores planned for 2026; 345 overseas stores; 7 overseas markets. (brie…	PUBLIC
L-gabgrowth	GabGrowth (secondary): first overseas store, Kuala Lumpur, reportedly sold 1,000+ cups on opening day. (brief.md section 9)	SECONDARY
L-unsw	UNSW BusinessThink, Oct 2024: Chagee semi-automation, baristas scan a QR code and the machine dispenses the formulation. (brief.md section 9)	PUBLIC
L-mom-pwm	Ministry of Manpower, Progressive Wage Model, Food Services sector, https://www.mom.gov.sg/employment-practices/progressive-wage-model/food-services-sector (page u…	PUBLIC
L-cpf	CPF Board, 'How much CPF contributions to pay', https://www.cpf.gov.sg/employer/employer-obligations/how-much-cpf-contributions-to-pay (page updated 11 Aug 2026; f…	PUBLIC
L-mom-ows	MOM Occupational Wages 2025, Table 1 (median monthly basic and gross wages of full-time resident employees, June 2025; released 30 Jun 2026), https://stats.mom.gov…	PUBLIC
L-sdl	CPF Board, Skills Development Levy, https://www.cpf.gov.sg/employer/employer-obligations/skills-development-levy (updated 2 Jul 2026): 0.25% of monthly total wages…	PUBLIC
L-mom-leave	MOM, Annual leave eligibility and entitlement, https://www.mom.gov.sg/employment-practices/leave/annual-leave/eligibility-and-entitlement (updated 17 Apr 2023): 7 …	PUBLIC
L-mom-ph	MOM, Public holidays entitlement and pay, https://www.mom.gov.sg/employment-practices/public-holidays-entitlement-and-pay (updated 30 Jul 2025): 11 paid public hol…	PUBLIC
L-prn	PR Newswire, 21 May 2024 (company release): Chagee says it sources all tea and milk directly for domestic operations, with distribution managed through HQ. (brief.…	PUBLIC
L-prometheus	Prometheus Capital Substack (SECONDARY - flag when used): own Yunnan plantations, cold chain to 6,000+ stores; tea lattes about 91% of China sales. (brief.md secti…	SECONDARY
L-chainreact	Chainreact Substack (secondary): founder avoided fruit tea partly for supply chain complexity. (brief.md section 9)	SECONDARY
L-triunity	Studio Triunity, citing the Mixue prospectus: Chagee's pure milk-tea menu seen as simpler to export; Mixue has 7 SEA warehouses and 3 planned local factories. (bri…	PUBLIC
L-cyberrt	Cyber-RT / Citizens Journal, Oct 2022: Chagee Express (Malaysia), semi-automated machine, three drink categories for speed. (brief.md section 9)	PUBLIC
L-gongcha	FastCasual and franchise.org, Jan-Feb 2026 (company-reported): Gong cha Super Wu, up to 65% higher peak-hour productivity, about one minute off prep, stores with a…	PUBLIC
L-starbucks	Constellation Research; Harvard D3: Starbucks Deep Brew - forecasting, scheduling, replenishment, order sequencing; location analytics for new stores. (brief.md se…	PUBLIC
L-mosimtec	MOSIMTEC, Winter Simulation Conference 2025: store digital twins replicate existing layouts. (brief.md section 9)	PUBLIC
L-slate	Slate, Aug 2021: bubble tea shaker robot needed about one worker vs four to six; two earlier all-robot shops closed within about a year. (brief.md section 9)	PUBLIC
I-leeyen	INFORMAL - call with Lee Yen. Do not cite without checking with her: Yunnan standard sourcing, GM-led new-market forecasting, app-only ordering with tourist except…	INFORMAL
L-xrates	X-Rates monthly averages, https://x-rates.com/average/?amount=1&from=USD&to=SGD&year=2026 (fetched 4 Oct 2026). September 2026 USD to SGD average 1.272467 (30 days…	SECONDARY

Claims we did not make

- That Chagee is first at anything, or that anything is not available in the market.
- Any revenue-mix share for franchise supply: sources conflict.
- Overseas milk sourcing details: unknown, so milk costs are generic.
- A Supply Chain Security Act local-sourcing rule: low-quality source.
- Any Chagee internal cost, forecast method or automation plan not in the ledger.
- Measured serve times: every time on the store-replica slide is modelled; the observed row is empty.
- That member data narrows the forecast band: the narrowing is our assumption, tested as a sensitivity.
- That all-human or mixed is proven cheaper: the gap is small and rests on a simplified staffing model.
- A saving against today's practice: the benefit is a modelled avoided cost against a standard counter.
- Anything about menu prices: that would need a demand-elasticity assumption we do not have.
- The informal sub-one-minute fulfilment target from the industry call: not used.
- That the prototype contains a trained AI model: it calculates from synthetic, assumed inputs.
- That rerunning the model with observed inputs validates it, or proves machines should be added.
- That uncertainty never matters for the counter: the finding holds for the scenarios we tested.
- Sales revenue or a payback for Chagee: A4 sets modelled avoided cost against approved but illustrative costs.
- Quantities, lead times, shelf lives, minimum orders or origin for tea, other ingredients or packaging.
- Inventory savings beyond fresh milk: tea, other ingredients and packaging have not been estimated.
- That inventory and counter are optimised together: the prototype treats them separately.

Over 5 years the modelled savings cover the cost of building and running the gates in year 5, leaving about S$289k ahead (illustrative)

ITEM	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	ASSUMPTION / HOW CALCULATED
Benefit: modelled avoided cost (no sales revenue)						
New stores opened	200	200	200	200	200	About 200 overseas openings planned for 2026 (L-jiemian); held flat after that (no later figure in our sources)
Store-years using the gates	100	300	500	700	900	Earlier stores trade all year; new stores 50% of a year (mid-year openings)
Counter saving (recurring)	S$228k	S$683k	S$1,138k	S$1,594k	S$2,049k	Store-years × S$2,277: the mixed counter's yearly cost above the cheapest counter, openings spread over our volume range (slide 9)
Fresh-milk opening stock (once per opening)	S$74k	S$74k	S$74k	S$74k	S$74k	Openings × S$370: best milk order with no signal minus with the member signal; if the signal narrows the range as assumed. Tea, other ingredients and packaging: not estimated
Total benefit	**S$302k**	**S$757k**	**S$1,213k**	**S$1,668k**	**S$2,123k**	
Cost of the initiative: one-time						
Build the four gates	S$504k	–	–	–	–	4 people × 6 months × 21 days × S$1,000 a day: forecast, ordering, counter comparison, review
Data integration and set-up	S$250k	–	–	–	–	POS sales, app sign-ups and supplier order feeds; lump sum
Total one-time cost	**S$754k**	**–**	**–**	**–**	**–**	
Cost of the initiative: recurring						
Maintenance team	S$504k	S$504k	S$504k	S$504k	S$504k	2 people × 12 months × 21 days × S$1,000; counted from year 1 (conservative)
Platform and licences	S$60k	S$180k	S$300k	S$420k	S$540k	Store-years × S$50 a month × 12
Gate 4 reviews	S$200k	S$200k	S$200k	S$200k	S$200k	Openings × 1 analyst day × S$1,000: updated scenario against the original plan
Total recurring cost	**S$764k**	**S$884k**	**S$1,004k**	**S$1,124k**	**S$1,244k**	
Total cost	**S$1,518k**	**S$884k**	**S$1,004k**	**S$1,124k**	**S$1,244k**	
Net benefit	**(S$1,216k)**	**(S$127k)**	**S$209k**	**S$544k**	**S$879k**	**Total benefit minus total cost**
Cumulative net	**(S$1,216k)**	**(S$1,343k)**	**(S$1,135k)**	**(S$591k)**	**S$289k**	**From year 1**

Sensitivity, cumulative net after 5 years. one-time costs double: S$465k short, no payback; counter saving halves: S$2,558k short, no payback; openings fall to 100 a year: S$1,493k short, no payback.

Read with care. No sales revenue: we have no price or demand-elasticity assumption, so the benefit is the modelled avoided cost from slide 9, against a mixed counter at every opening; it is not a demonstrated AI saving. The cost lines are our approved illustrative assumptions (A1, business case), not Chagee or vendor figures. Values in S$ thousands; brackets are negative.

In the peak hour, robots hand over a cup in about 49 seconds; a counter like today's takes about 2.6 minutes

INTERACTIVE EXPLAINER STORE REPLICA: TWO COUNTERS IN THE PEAK HOUR, SIDE BY SIDE

Illustrative layout after a Chagee tea bar: order right, brewing behind the counter, pick-up left. Choose each counter; people move as the Week 5 bucket brigade; staff and arrivals from the cost model.



Number = worker speed · cup above = drink in hand · faded = walking back · red ring = held back by the colleague ahead · ▲ = handover. Robots stay put; cups move. Both counters see the same customers.

Printed at minute 20 of the peak hour, moderate surge: people slowest first (top) against fastest first (bottom).

Time to serve, cost model (median · slowest tenth · lost):

Today's counter **2:34** · 5:03 · 2% lost All-human **2:42** · 5:10 · 2% lost All-robot **0:49** · 1:19 · 0% lost

Observed: not measured yet

OUR VIEW

Robots buy speed in the peak; whether it pays is the counter-cost slide.